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Designing a DAB using Scaling Laws

Reference:

https://ridleyengineering.com/images/webinar/DAB_Design_Using_Scaling_Laws.pdf

Aims:

LV stage:

$V_{nom} = 48 \text{ V}$

$V_{min} = 24 \text{ V}$

$V_{max} = 60 \text{ V}$

$P_{nominal} = 5 \text{ kW}$

$I_{max} = 104 \text{ A}$

HV stage:

$V_{nom} = 400 \text{ V}$

$V_{max} = 420 \text{ V}$

$V_{min} = 360 \text{ V}$

$P_{nominal} = 5 \text{ kW}$

$I_{max} = 12.5 \text{ A}$

Fsw starting point = 100 kHz

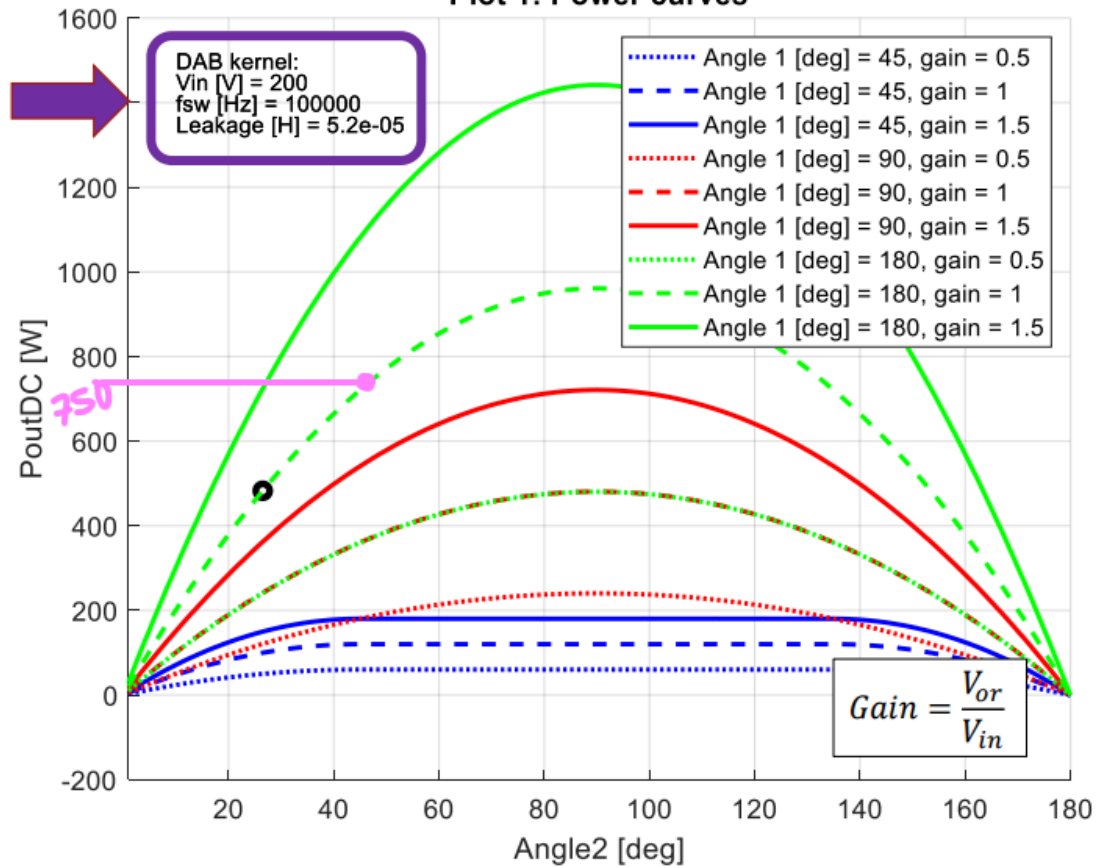
Choose a Single Phase Shift mode

I.e. (Intra bridge) Angle 1 = 180°

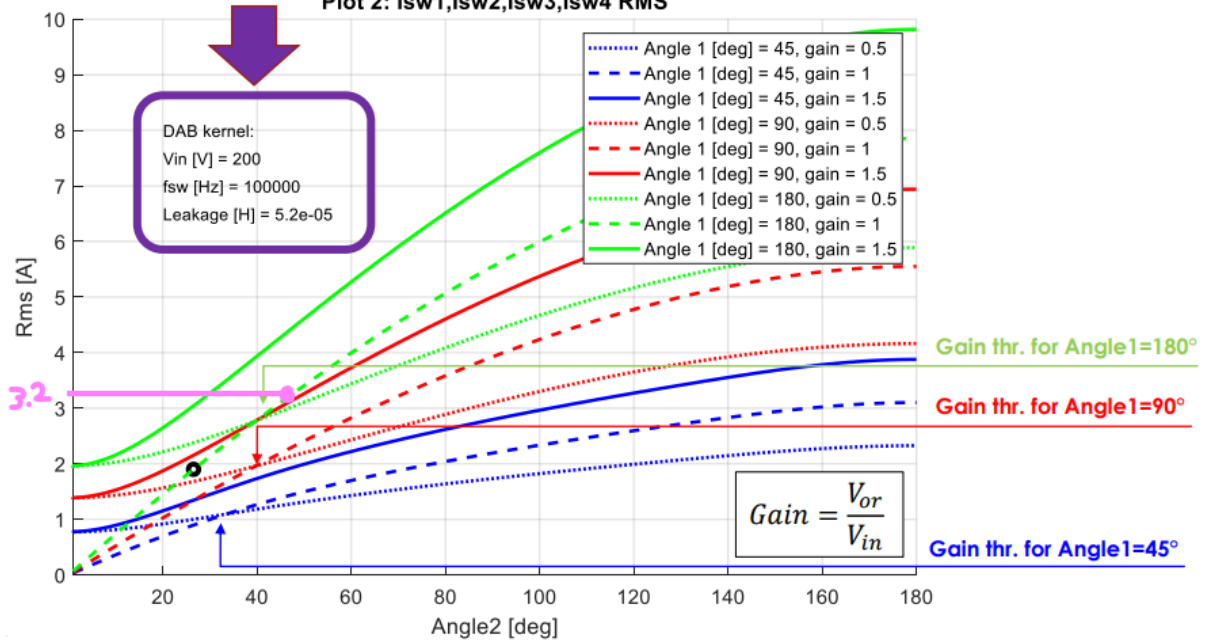
(Inter bridge) Angle 2 = Control angle up to 45°

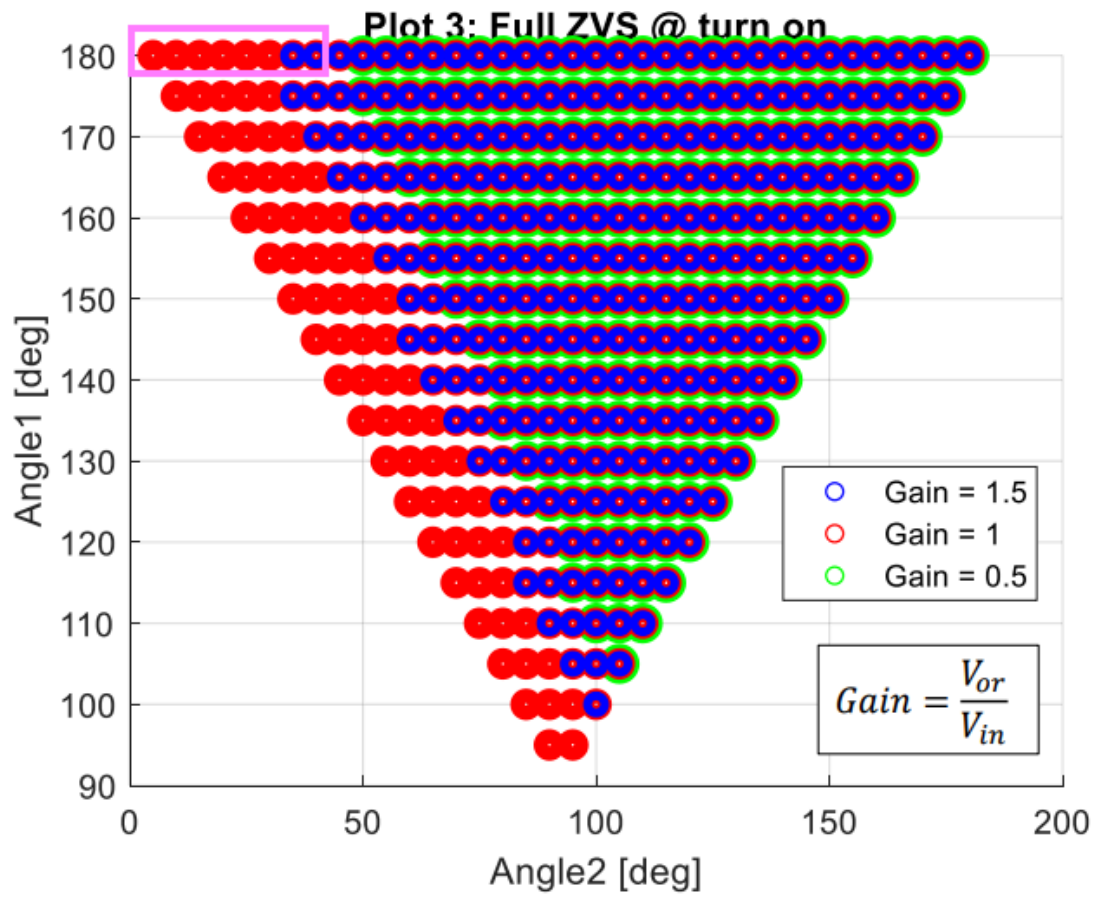
Gain = 1

Plot 1: Power curves



Plot 2: Isw1, Isw2, Isw3, Isw4 RMS





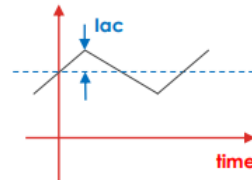
The Scaling laws: « three is a magic number! »

- To double the power at a certain frequency (keeping all else unchanged such as the phase angles), we need to halve the inductance and double the capacitance.
- To double the frequency, keeping the same power, once again we just need to halve the inductance and halve the capacitance. Recalling the buck converter equations.

$$L = \frac{V_{out} \cdot (1-D)}{r \cdot f_{sw} \cdot I_{out}}$$

$$C = \frac{I_{out} \cdot r}{8 \cdot f_{sw} \cdot \Delta V}$$

$$r = \frac{2 \cdot I_{ac}}{I_{dc}}$$



- Added guest to the party: If we double the input voltage, we will quadruple the power. Proof below:**

If:

$$\triangleright P_{out} \cong P_{in}$$

$$\triangleright V_{out} = TF_{dc} \cdot V_{in}$$

then:

$$\triangleright P_{out} = \frac{V_{out}^2}{R_{load}} = \frac{(TF_{dc} \cdot V_{in})^2}{R_{load}}$$

If Vin doubles, to keep the equation valid, the power shall quadruple!

3

DAB Design 1

Pout = 7 kW, Vout = 48V, Vin = 380V, fsw = 100kHz

Assumptions

- Best entry Angles are Angle 1 = 180° and Angle 2 ≈ 26°
- Best gain entry point is: gain = 1

Before: calculate the scaling factors!

$$k_{voltage} = \frac{V_{in\,eff}}{V_{kernel}} = \frac{380}{200} = 1,9 \rightarrow \text{Power boost} = 1,9^2$$

$$k_{power} = \frac{P_{eff}}{P_{kernel}} = \frac{7k}{480 \cdot 1,9^2} = 4$$

$$k_{frequency} = \frac{f_{eff}}{f_{kernel}} = \frac{100k}{100k} = 1$$

Scaling from the kernel to the new requirements

One Page DAB design

After: calculate the Leakage inductance, the turn ratio and both capacitors):

$$L_{Lk} = \frac{L_{kernel}}{k_f \cdot k_p} = \frac{52\, \mu H}{4} = 12,8\, \mu H$$

$$C_{dc\,block} \approx 100 \cdot \frac{1}{4 \cdot \pi^2 \cdot f^2 \cdot L_{lk}} \approx 20\, \mu$$

$$N = \frac{N_p}{N_s} = \frac{380}{48} = 7.91$$

$$C_{out} \approx \left(I_{Llk\,peak} \cdot \frac{N_p}{N_s} - \frac{P_{out}}{V_{out}} \right) \cdot \frac{T}{\Delta V_{out\,(1V)}} \approx 110\, \mu$$

48

Kernel power: 750 W (Angle 1 = 180 °, Angle 2 = 45 °, Gain = 1)

Kernel Fsw = 100 kHz

Kernel Leakage = 5.2e-05 H

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Kernel Input voltage: 200 V

$I_{Llk_peak} = 10 \text{ A}$

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HV stage:

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$V_{min} = 360 \text{ V}$

$P_{nominal} = 5 \text{ kW}$

$I_{max} = 12.5 \text{ A}$

Fsw starting point = 100 kHz

$T = 1 / 100 \text{ kHz} = 10 \text{ us}$

Tangle2 = 1.25 us

Choose a Single Phase Shift mode

I.e. (Intra bridge) Angle 1 = 180 °

(Inter bridge) Angle 2 = Control angle up to 45 °

Gain = 1

$k_{voltage} = V_{in_eff} / V_{kernel} = 48 / 200 = 0.24 \rightarrow \text{Power "boost"} 0.24^2 = 0.058$

$k_{power} = P_{eff} / P_{kernel} = 5000 / (750 * 0.24^2) = 115.741$

$k_{frequency} = f_{eff} / f_{kernel} = 100 \text{ k} / 100 \text{ k} = 1$

$L_{lk} = L_{kernel} / (k_{frequency} \times k_{power}) = 52 \text{ uH} / (1 * 115.741) = 0.450 \text{ uH}$

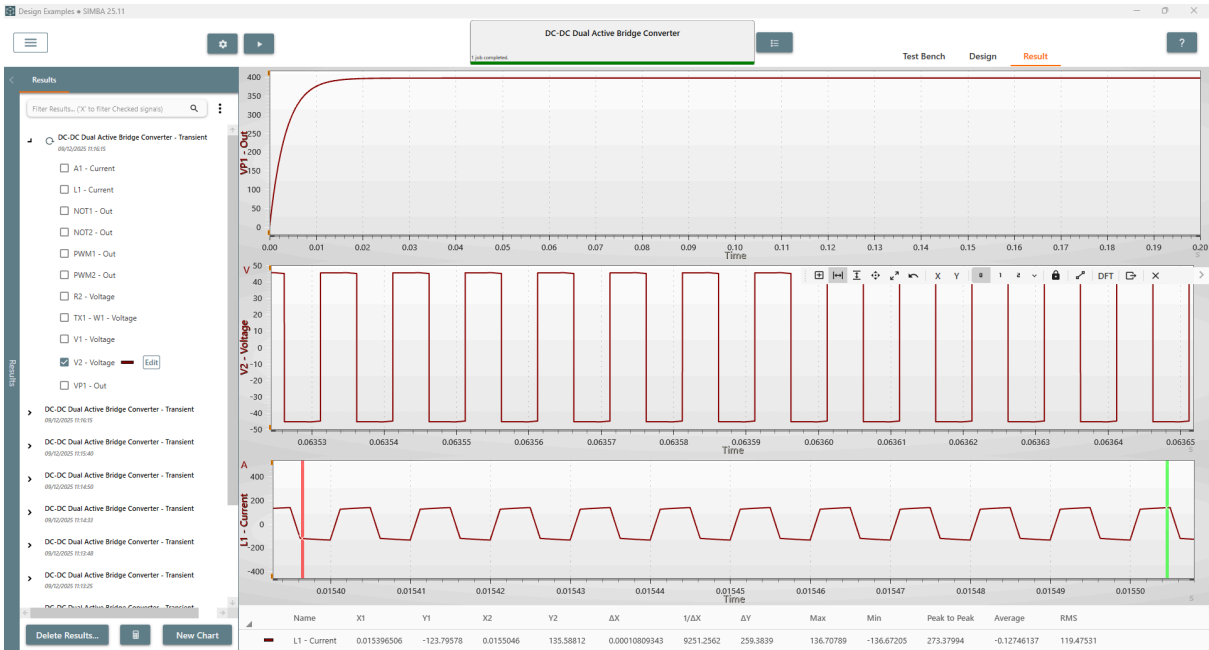
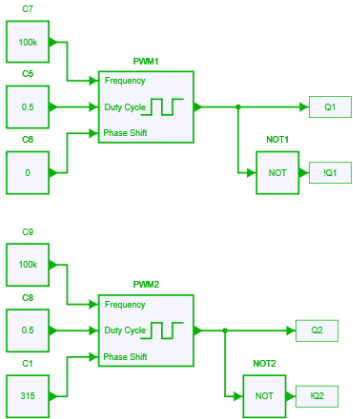
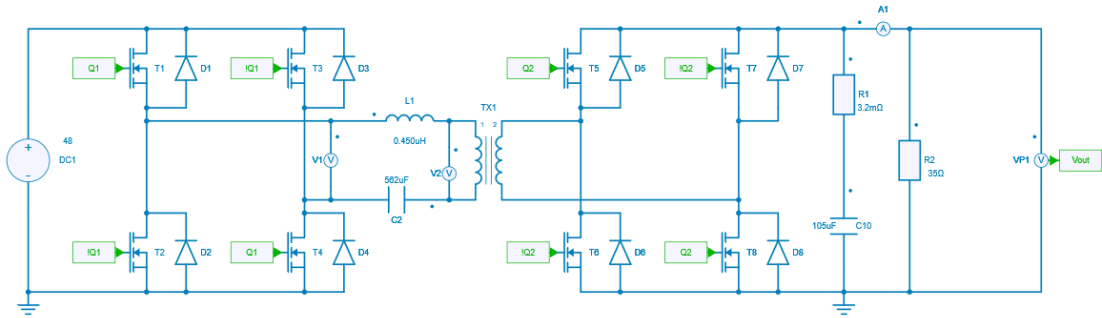
$C_{dcblock} = 100 \times (1 / (4 \times \pi^2 \times f^2 \times L_{lk})) = 562 \text{ uF}$

$N = N_p / N_s = 48 / 420 = 0.114$

$I_{Llk_peak} = (I_{Llk_peak_kernel} \times k_{voltage}) \times (P_{eff} / P_{kernel}) = (10 \times 0.24) / (5000 / 750) = 277.77 \text{ A}$

$C_{out} = (I_{Llk_peak} \times (N - P_{out} / V_{out})) \times ((T/2 - Tangle2) / \Delta V_{out}) = (277.77 \times 0.114 - 5000/420) \times ((10\text{u}/2 - 1.25\text{u}) / 1) = 105 \text{ uF}$

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Validation:

396 V on 35 Ohm resistor $\rightarrow P = V^2 / R = (396^2) / 35 = 4480.457 \text{ W}$

Note that 50 mOhm R_{ds_on} MOSFETs resulted in a significant voltage drop. Changed to 5 mOhm and this is the result.

Voltage ripple $\sim 0.2 \text{ V}$

Input into openmagetics:

Core: <https://uk.rs-online.com/web/p/transformer-cores/6479389>

Coil former: <https://uk.rs-online.com/web/p/transformer-accessories/1253682>

Windings: <https://www.rmcybernetics.com/shop/wire-cable/litz-wire>

“40A” and “6A” variants for primary and secondary respectively. Should be possible to wind to get $\sim 450 \text{ nH}$ leakage

Costs:

- Core = £18.54
- Primary wire (1 m) = £18
- Secondary wire (10m) = £38
- Coil former = £10
- Total $\sim$ £85

Ranges from 293 nH to 1300 nH depending on winding arrangement.

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Steps

Design Req.

Op. Points

Magnetic Builder

Summary

Continue

Tool menu

Settings

Change tool

Shape Family

Shape

Manufacturer

Material

Number of Stacks

Gap info

Type

Length

Magnetizing Inductance

Core Losses

Saturation Proportion

E 70/33/32

TDK

Nb7

1

Ground

0 mm

322.33 uH

0.48 W

8.94 %

Advise core

Customize

Load core from library

Magnetic Builder

Error in wire

Plot current density

No

Yes

Winding 1

Winding 2

No. Turns

No. Parallels

Type

Standard

Cond. diameter

Coating

No. Strands

Eff. Current Density

Eff. Skin Depth

Outer Width

Outer Height

4

1

Litz

IEC 60317

0.1 mm

TiW, TR 155, BV 13.5 kV

2000

7.59 A/mm²

204.02 um

6.90 mm

6.90 mm

Advise wire

Customize

Load wire from library

Advise all

Customize

Load wire from library

Bobbin Wall Th.

Bobbin Column Th.

Section Interl. Order

Fill. F_{area}

Fill. F_{length}

Winding losses

Total losses

Leakage Inductance

1.827 mm

2.192 mm

12

74.5 %

97.8 %

33.97 W

34.45 W

480.18 nH

Show alignment options

Show insulation options

Customize

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Magnetic Summary

Core data

Core Shape

Name

No. Stacks

Effective length

Effective area

Effective volume

Minimum Area

Core Gapping

Type

Length

Height offset

Subtractive

Residual

Residual

Material Parameters

Name

Manufacturer

Permeance (AL value)

Initial Permeability (μ)

Eff. Permeability (μ_e)

Resistivity

Curie Temperature

Density

E 70/33/32

1

0.16 m

682.89 mm²

102.956.78 mm³

676.24 mm²

0 m

5 μm

5 μm

Nb7

TDK

25°C

100°C

12.25 μH/(m·T²)

20.15 μH/(m·T²)

2304

3983

2141

3520

10.2 m

210 °C

4.85 Mg/m³

Coil data

Coil Global Parameters

No. Sections

No. Layers

Primary

No. turns

No. parallels

Wire

Secondary

No. turns

No. parallels

Wire

4

6

4

1

Litz (undefined)

35

1

Litz (undefined)

Simulation result per Operating Point

Op. Point No. 1

Core

Mag. Ind.

Core losses

Core temp.

B peak

322.3 μH

0.48 W

102.08 °C

44.27 mT

Coil

Winding

DC Res.

Curr. Density

Wind. Loss

Link. Ind.

Primary

Secondary

1.58 mΩ

15.18 A/mm²

23.28 W

0.12 Ω

10.29 A/mm²

19.67 W

371.17 nH

Coil Losses Breakdown

Winding

Ohmic Loss

Skin Loss

Proc. Loss

Primary

Secondary

22.46 W

1.76 mW

0.81 W

14.72 W

3.12 mW

4.95 W

Core Coil

Show H field

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Approximately 44 W of losses to deal with on the E type design.

Magnetics iterations

Tried out both a Toroidal core and an ETD core.

The toroidal core option came in at low cost and low loss. 1.7 W core loss, 26 W of winding losses. BUT the winding and mounting arrangements required extra consideration. There was no off-the-shelf mount for the toroid and with Litz wire it wouldn't be self-supporting.

So the ETD option comes in with ~22 W of total losses after adjusting the number of windings slightly and the size of the secondary Litz wire. It's likely the losses of the other variants could have been reduced to these levels with some tweaking.

Litz vs non-Litz. At 100 kHz and these current levels, the difference between Litz and standard wires was significant! It made 10's of Watts of loss difference for similar size wires. Likely to cause some serious overheating problems, as well as the efficiency hit!

The plus side of using the ETD core (ETD 59/31/22) is also that you can buy an off-the-shelf winder and yoke (to clip the core to the winder). This is useful because it gives a really nice mounting arrangement that lets you directly insert the transformer into a PCB or connect up with wires for testing.

Protection thoughts

How to protect against overcurrent and overvoltage?

On the DC side, considering an input current of $> 100\text{ A}$ we're beyond the realms of resettable fuses.

I received some sage wisdom that in development an e-Fuse could be a very good investment of time and money. This is basically a very fast current sensor, comparator and full-rate switch put in series.

I plan to make a couple of these as development tools but likely won't be in the final product as it's an unnecessary set of parts. Although maybe it'll be great for test-ability repairability, who knows until we try it!

But then we should probably have a back-up fuse as the e-Fuse itself could fail. So we need a big, but standard, DC fuse on the input.

We probably also want fuses either side of our DC-link capacitors as they could be a significant overcurrent source in the case of a fault somewhere else in the circuit. It would be great if a fuse blows instead of the PCB itself.

Then on the AC side, connecting to the grid; I expect we'll need some din-rail mount gear similar to a small consumer unit. A 32 A, 30 mA RCBO is probably a good start and include the leakage current protection.

For over-voltage protection, there are some trade-offs to consider:

1. Do you actually need it?
2. How likely is the fault?
3. How bad is the damage of the fault?
4. How effective is the protection?

For example, at the DC we're expecting a battery or solar panels depending on the model. Early versions will be based on interfacing with a battery as this is easier to get started with. So for a battery, what if I put a Transient Voltage Suppressor (TVS) on the DC-DC converter board? Will it blow up in the case of an over-voltage before it does any good? Or can it be sized, in relation to the fuse, to limit overvoltage and blow the fuse before anything else goes pop? It probably can, but there's going to be some careful sizing needed.

Then a similar trade is needed either side of the 400 V DC-link. An overvoltage here is actually pretty likely because the DC-DC converter we're using is a DAB converter and the theoretical gain of that is quite large if the control system operates at a fixed phase shift and the load is high impedance. This would be bad news for our DC-link as the capacitors would definitely go pop if we exceed their voltage rating. So a TVS seems like a good idea! Making sure this protection will actually work, again, will need some careful sizing! The protection voltage needs to be sufficiently higher than the nominal working voltage to avoid the TVS (or MOV) taking current in normal operation.

On the AC connection side, this is a bit easier and a simple MOV is a no-brainer to account for surges from the grid.

We should also consider the real-time protection elements of protection. For a DC-DC converter this could be as simple a protective action as stopping the “Enable” line for the gate drivers - stopping all active switching. This should shut things down pretty quickly if everything is working as it should! Now why would we shut down the converter?

1. Over-temperature -> we should measure temperature directly and ensure nothing is getting too hot. This is ultimately why we do overcurrent protection in most cases, so a direct measurement is a great idea!
2. Instantaneous over-current through the MOSFETs -> This is a measurement we want to be taking for current-mode control anyway. So let's use it for some real-time protection to prevent things going pop up-front. This can be done as a simple comparator against a fixed value set for the installed MOSFETs. Ideally this would be done with separate hardware, but it still adds a lot of value when done in software.
3. Grid compliance -> There are various states defined in the different grid codes we want to comply with that mean you need to be able to switch off your inverter when those conditions are met. Generally related to voltage, power factor and frequency of the grid.

Passive assistance for protection:

1. Where possible, set the 0 of digital signals to the “safe” state. An example is the enable pin on the gate drivers. If power is lost to the enable pin, the switches won't be switch on. This is generally a safe state, especially if all the gate drivers are connected to the same enable pin.
2. Limit possible dV/dt and di/dt through filters. If we can slow down rates of change, it's easier to protect against them (in general). An example is to make sure there's enough inductance in and out of the DC-link capacitor bank to limit the rate of change of current in case of a short. Otherwise that short circuit may make things go pop before a fuse even has a chance to act! This needs to be considered as a trade though, too much inductance and the necessary di/dt swings for normal operation will generate excessive voltage spikes!